ENGINEERING TRIPOS PART IIA

ELECTRICAL AND INFORMATION ENGINEERING
TEACHING LABORATORY

MODULE EXPERIMENT 3F1

FLIGHT CONTROL

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Objectives:

- Simulation of various aircraft models on the computer.
- Study real-time (manual) control and the limitations imposed by time delays.
- Design of a simple autopilot.
- Illustrate frequency response concepts in analogue and digital control systems, conditions for oscillation in feedback systems and stability.
- Gain familiarity with Python and Jupyter Notebooks.

3F1 Lab Worksheet

2.1 Simplified Aircraft Model

Transfer function $= \dfrac{10}{s^2 + 10s}$

`num = [10]`   `den = [1, 10, 0]`

2.2 Modelling Manual Control

Controller transfer function $= K(s) = ke^{-sD}$

$k = 0.8$ $D = 1$ s

Phase margin $= 45$ degrees

Amount of extra time delay which can be tolerated $= \dfrac{45 \times \pi/180}{0.8} \approx 0.98$ s (approx 1.0 s)

2.3 Pilot Induced Oscillation

Period of oscillation (observed) $= 5$ s

Period of oscillation (theoretical) $= \dfrac{2\pi}{\omega_{pc}} = \dfrac{2\pi}{0.8} \approx 7.85$ s

2.4 Sinusoidal disturbances

Maximum stabilising gain $= 4.5$ (Gain Margin)

Gain at 0.66 Hz $= 5.5$ dB   Phase at 0.66 Hz $= -90$ degrees

2.5 Unstable Aircraft

Fastest pole at $T = 0.6$

3.2 Autopilot with Proportional Control

Proportional gain $K_c = 12$ Period of oscillation $T_c = 2$ s

3.3 Autopilot with PID Control

Transfer function of PID controller $= K_p \left(1 + \dfrac{1}{T_i s} + T_d s\right)$

PID constants: $K_p = 7.2$ $T_i = 1.0$ $T_d = 0.25$

Adjusted value of $T_d = 0.35$

3.4 Integrator Wind-up

Integrator bound $Q = \dfrac{d \cdot T_i}{K_p} = \dfrac{2 \times 1.0}{7.2} \approx 0.278$

3F1 Flight Control Lab Report

Question 1. *(§2.2 Modelling Manual Control) Nyquist diagram (from Bode diagram) for controller in series with plant.*

System Model: The simplified aircraft dynamics satisfy

$$\ddot{y}(t) + M\dot{y}(t) = Nx(t)$$

Taking the Laplace transform (with zero initial conditions), the plant transfer function is:

$$G(s) = \frac{N}{s^2 + Ms} = \frac{N}{s(s+M)}$$

This represents an integrator ($1/s$) in series with a first-order lag ($1/(s+M)$). The human controller is modelled as a proportional gain k with pure time delay D:

$$K(s) = ke^{-sD}$$

Open-Loop Transfer Function: The loop transfer function is:

$$L(s) = K(s)G(s) = \frac{kN \cdot e^{-sD}}{s(s+M)}$$

Frequency Response Analysis: Substituting $s = j\omega$:

$$L(j\omega) = \frac{kN \cdot e^{-j\omega D}}{j\omega(j\omega + M)}$$

The denominator can be written as:

$$j\omega(j\omega + M) = j\omega \cdot |j\omega + M| \cdot e^{j\arctan(\omega/M)} = \omega\sqrt{\omega^2 + M^2} \cdot e^{j(\pi/2 + \arctan(\omega/M))}$$

Therefore, the magnitude and phase are:

$$|L(j\omega)| = \frac{kN}{\omega\sqrt{\omega^2 + M^2}}, \quad \angle L(j\omega) = -\frac{\pi}{2} - \arctan\left(\frac{\omega}{M}\right) - \omega D$$

Asymptotic Behaviour:

- **Low frequency** ($\omega \to 0$): $|L| \to \infty$, $\angle L \to -\pi/2$ (integrator dominates).
- **High frequency** ($\omega \to \infty$): $|L| \to 0$, $\angle L \to -\infty$ (time delay adds unbounded phase lag $-\omega D$).

Substituting experimental values ($M = N = 10$, $k = 0.8$, $D = 1$s):

$$L(j\omega) = \frac{8e^{-j\omega}}{j\omega(j\omega + 10)}$$

From the Bode diagram (Figure 2), the gain crossover frequency $\omega_{gc} \approx 0.8$ rad/s. At this frequency:

$$\angle L(j \cdot 0.8) = -90^\circ - \arctan(0.08) - 0.8 \times \frac{180^\circ}{\pi} \approx -90^\circ - 4.6^\circ - 45.8^\circ \approx -140^\circ$$

The phase margin is PM $= 180° - 140° = 40°$ (approximately 45° from graph).

Constructing Nyquist from Bode: The Nyquist plot traces $L(j\omega)$ as ω varies from 0 to ∞. Starting from the negative imaginary axis (phase $-90°$, large magnitude), the locus spirals clockwise towards the origin. The critical point -1 is not encircled, confirming closed-loop stability.

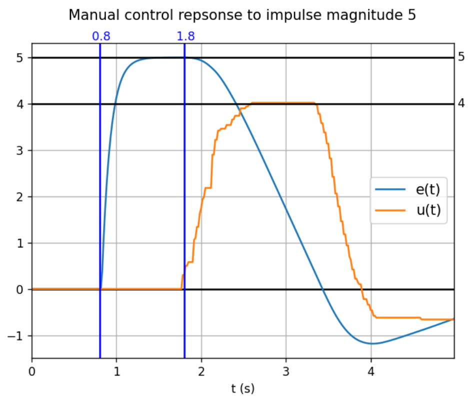


Figure 1: Manual control response to impulse disturbance. From the plot: $k \approx u_{max}/e_{max} = 4/5 = 0.8$, $D \approx 1$s (delay before response).

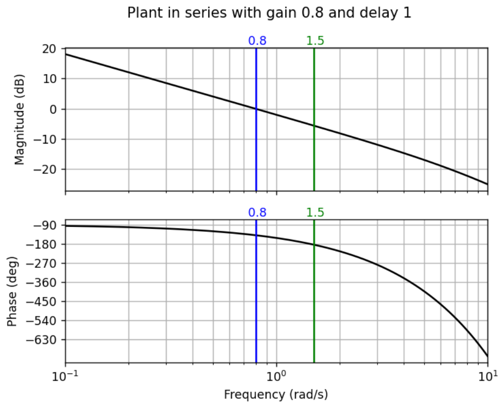
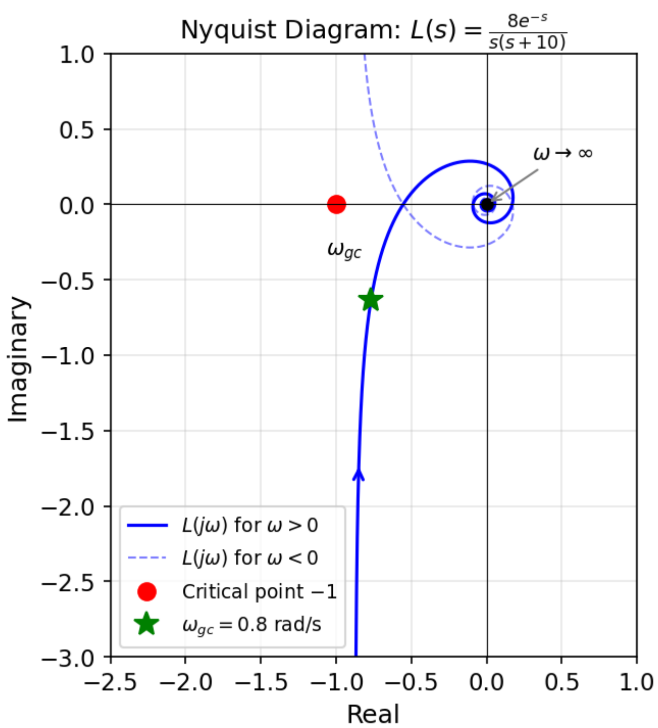


Figure 2: Bode diagram: open-loop $L(s)$ with $k = 0.8$, $D = 1$s. Phase margin $\approx 45°$ at $\omega_{gc} = 0.8$ rad/s.



Nyquist diagram of $L(s) = K(s)G(s)$. The locus spirals clockwise from the negative imaginary axis toward the origin. The critical point -1 (red) is not encircled.

2. (§2.2 Modelling Manual Control) Are you using any integral action?

Yes. In Figure 3, the manual control input $u(t)$ does not return to zero but maintains a steady value to counteract the step disturbance, while $e(t) \to 0$. This implies the human controller provides integral action ("trimming") to eliminate steady-state error. The simple model $K(s) = ke^{-sD}$ is therefore an approximation that neglects the pilot's low-frequency adaptation.

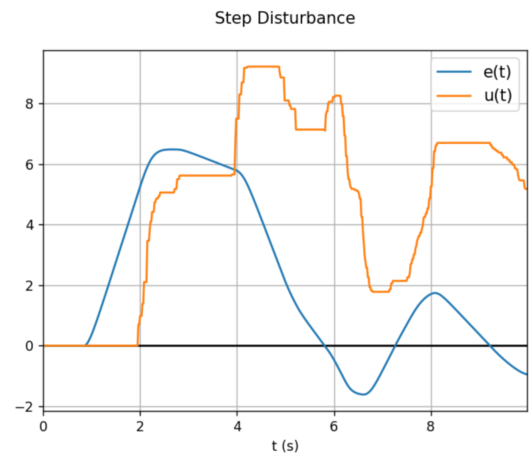


Figure 3: Step disturbance response showing integral action: $u(t)$ maintains steady offset to drive $e(t) \to 0$.

3. (§2.3 Pilot Induced Oscillation) Explain the oscillation of the feedback loop.

Mechanism: PIO occurs when the pilot's time delay introduces sufficient phase lag that the total loop phase reaches $-180°$ at a frequency where the loop gain exceeds unity. The PIO aircraft has $G_1(s) = c/(Ts+1)^3$ which contributes up to $-270°$ phase lag, plus the pilot delay e^{-sD} adds $-\omega D$ radians.

Stability condition: At the phase crossover frequency ω_{pc} where $\angle L(j\omega_{pc}) = -180°$, if $|L(j\omega_{pc})| > 1$, the system is unstable and oscillates at frequency ω_{pc}.

Period comparison: From Bode (Figure 6), $\omega_{pc} \approx 0.8$ rad/s, giving theoretical period $T = 2\pi/\omega_{pc} \approx 7.85$s. The observed period ($\approx$ 5s) is shorter, suggesting the pilot adapts their gain/delay under oscillatory conditions.

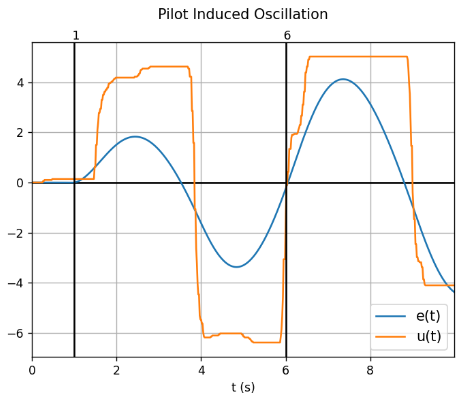


Figure 4: PIO with manual control. Period $\approx$ 5s.

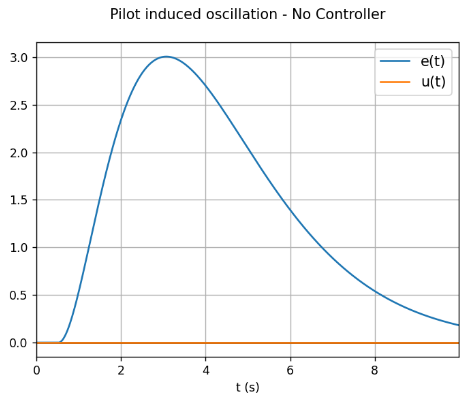


Figure 5: No control: aircraft is naturally stable.

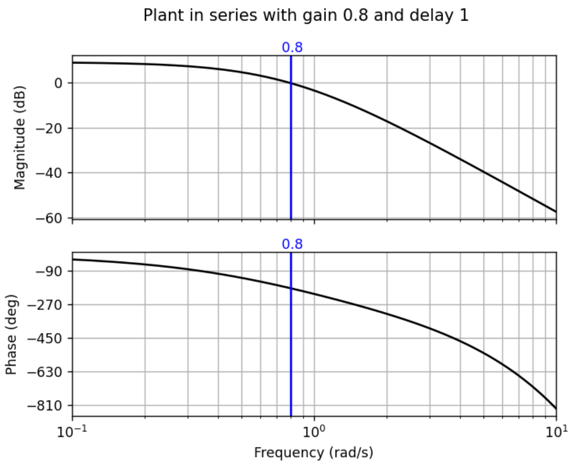


Figure 6: PIO Bode diagram. Phase crossover at $\omega_{pc} \approx 0.8$ rad/s.

4. (§2.3 Pilot Induced Oscillation) Can you give a rough guideline to the control designer to make PIO less likely?

To prevent PIO, ensure adequate phase margin at frequencies where the pilot is active ($<$ 1 Hz):

- Reduce total system delay (sensors, actuators, displays)
- Avoid plants with high phase lag at low frequencies

- Design for phase margin $> 45°$ at the expected pilot crossover frequency
- Consider rate limiters to prevent high-frequency pilot inputs

5. (§2.4 Sinusoidal Disturbances) Was your manual input able to reduce the error?

No. The closed-loop response to disturbance d is $y = \dfrac{G}{1+KG}d$. For disturbance rejection, we need $|1+KG| > 1$.

At 0.66 Hz ($\omega = 4.15$ rad/s), pilot delay adds phase $-\omega D = -4.15 \times 1 \approx -238°$. Combined with plant phase ($\approx -90°$ from Bode), total phase approaches $-330°$, meaning KG points nearly opposite to -1. This makes $|1+KG| < 1$, so the disturbance is actually *amplified* rather than attenuated. Manual control worsens the oscillation.

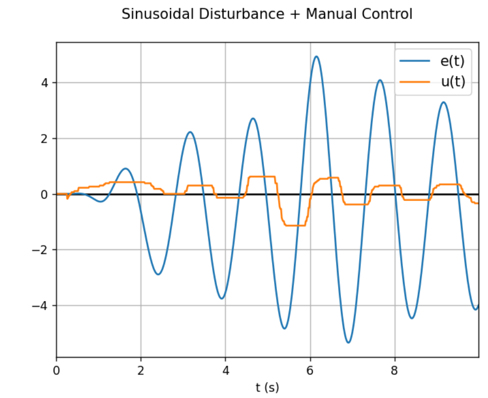


Figure 7: Sinusoidal + manual control.

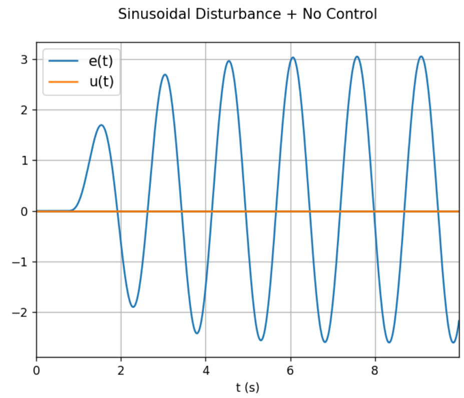


Figure 8: Sinusoidal + no control.

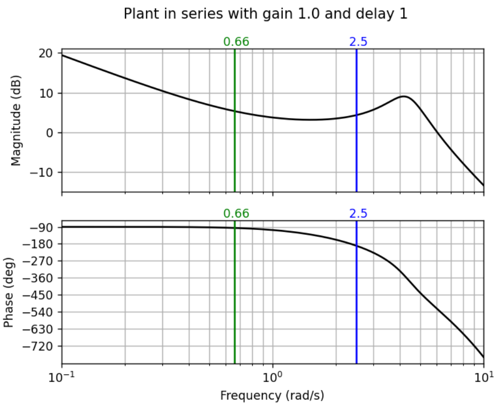


Figure 9: F4E Bode: gain margin 4.5; at 0.66 Hz: 5.5 dB, $-90°$.

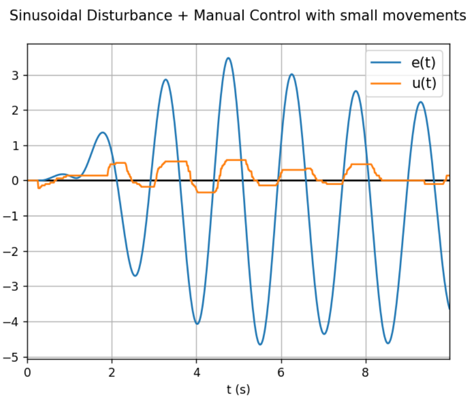


Figure 10: Small movements: still ineffective.

6. (§2.5 An Unstable Aircraft) Nyquist diagram for $G_2(s)$.

The unstable aircraft has transfer function:

$$G_2(s) = \frac{2}{sT - 1}$$

with one unstable (RHP) pole at $s = 1/T > 0$.

Frequency Response Derivation: Substituting $s = j\omega$:

$$G_2(j\omega) = \frac{2}{j\omega T - 1}$$

To find real and imaginary parts, multiply by conjugate:

$$G_2(j\omega) = \frac{2}{-(1 - j\omega T)} = \frac{-2(1 + j\omega T)}{(1 - j\omega T)(1 + j\omega T)} = \frac{-2(1 + j\omega T)}{1 + \omega^2 T^2}$$

Therefore:

$$\mathrm{Re}[G_2(j\omega)] = \frac{-2}{1 + \omega^2 T^2}, \quad \mathrm{Im}[G_2(j\omega)] = \frac{-2\omega T}{1 + \omega^2 T^2}$$

Geometric Interpretation: Letting $x = \mathrm{Re}$ and $y = \mathrm{Im}$:

$$(x + 1)^2 + y^2 = \left(\frac{-2}{1 + \omega^2 T^2} + 1\right)^2 + \left(\frac{-2\omega T}{1 + \omega^2 T^2}\right)^2 = 1$$

This is a circle centered at $(-1, 0)$ with radius 1.

Boundary Points:

- At $\omega = 0$: $G_2(0) = -2$ (leftmost point on real axis)
- As $\omega \to \infty$: $G_2 \to 0$ (approaches origin)

The locus is traced clockwise as ω increases from 0 to ∞.

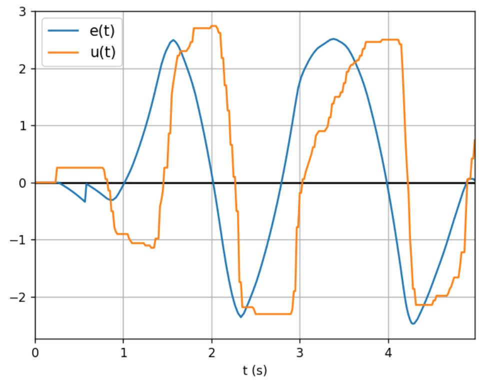
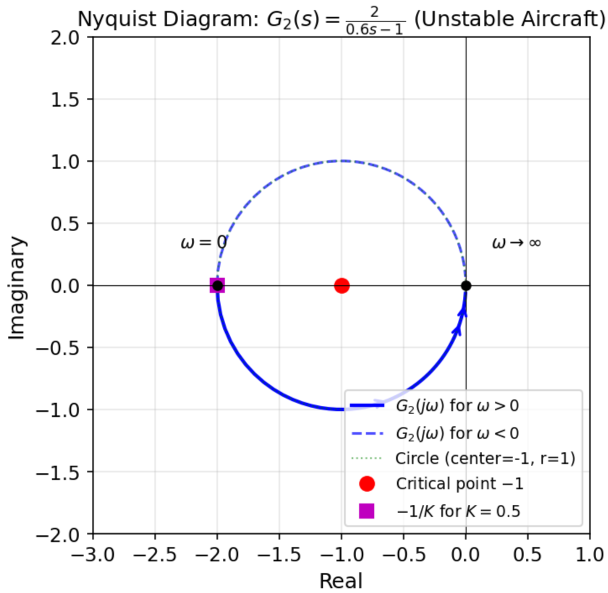


Figure 11: Unstable aircraft ($T = 0.6$) manually stabilized. Oscillatory near stability limit.



Nyquist diagram of $G_2(s) = 2/(0.6s - 1)$. Circle centered at $(-1, 0)$ with radius 1.

7. (§2.5 An Unstable Aircraft) Explain, using the Nyquist criterion, why the feedback system is stable with a proportional gain greater than 0.5.

Nyquist Criterion: For a system with P open-loop poles in the RHP, closed-loop stability requires $N = -P$ encirclements of $-1/K$, where $N > 0$ is counterclockwise.

Here $P = 1$ (one unstable pole), so we need $N = -1$: one *clockwise* encirclement of $-1/K$.

The Nyquist locus is a circle from -2 to 0 on the real axis, traversed clockwise. For $-1/K$ to be encircled clockwise:

$$-2 < -\frac{1}{K} < 0 \implies 0 < \frac{1}{K} < 2 \implies K > 0.5$$

For $K > 0.5$, the point $-1/K$ lies inside the circle, giving the required clockwise encirclement for stability.

8. (§2.5 An Unstable Aircraft) Sketch of a Nyquist diagram for $G_2(s)$ with a small time delay D.

With delay, the transfer function becomes $G_2(s)e^{-sD}$. The frequency response is:

$$G_2(j\omega)e^{-j\omega D} = \frac{2}{j\omega T - 1} \cdot e^{-j\omega D}$$

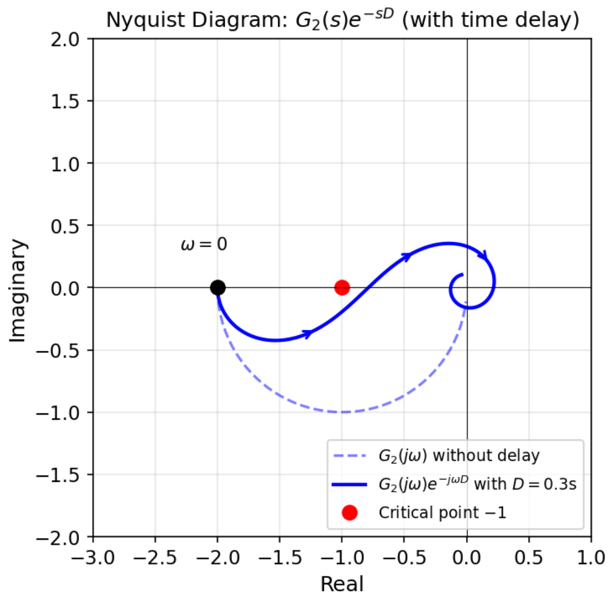
The delay contributes phase $-\omega D$ without changing magnitude:

$$|G_2(j\omega)e^{-j\omega D}| = |G_2(j\omega)|, \quad \angle(G_2 e^{-j\omega D}) = \angle G_2 - \omega D$$

Effect on Nyquist Locus:

- At $\omega = 0$: No change; still starts at $G_2(0) = -2$.
- As ω increases: Additional clockwise rotation by ωD radians.
- The circular locus becomes a spiral curving inward toward the origin.

For large D, the spiral may cross $-1/K$ multiple times, potentially destabilizing the system. This explains why manual stabilization becomes impossible when pilot delay D exceeds the system time constant T.



Nyquist diagram with time delay. Dashed: original $G_2(j\omega)$. Solid: with delay $e^{-j\omega D}$. The spiral deviates from the circle due to phase lag.

9. (§3.4 Integrator Wind-up) Explain how you calculated the bound on Q.

PID Controller: $u(t) = K_p \left(e + \frac{1}{T_i} \int_0^t e \, d\tau + T_d \frac{de}{dt} \right)$

Steady-state analysis: After settling, $e_{ss} = 0$ and $\dot{e} = 0$. To cancel a step disturbance d, we need $u_{ss} = d$. Only the integral term contributes:

$$u_{ss} = \frac{K_p}{T_i} I_{ss} = d \implies I_{ss} = \frac{d \cdot T_i}{K_p}$$

Minimum bound: For $d = 2$, $T_i = 1.0$, $K_p = 7.2$:

$$Q = \frac{2 \times 1.0}{7.2} = \frac{5}{18} \approx 0.278$$

Setting Q smaller would prevent full disturbance rejection (steady-state error). Setting Q larger allows more wind-up overshoot.

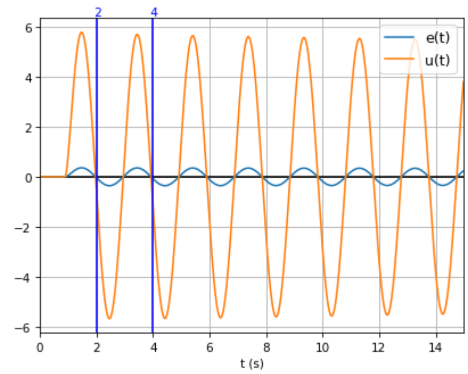


Figure 12: Proportional control: $K_c = 12$, $T_c = 2$s.

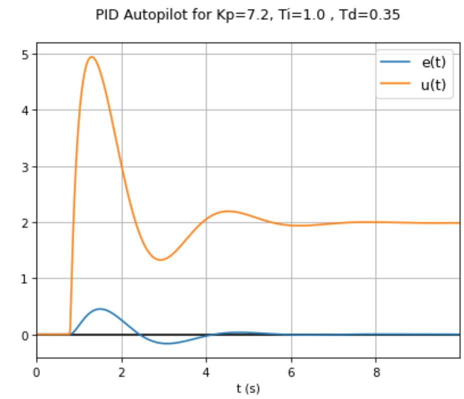


Figure 13: PID with $T_d = 0.35$: reduced oscillation.

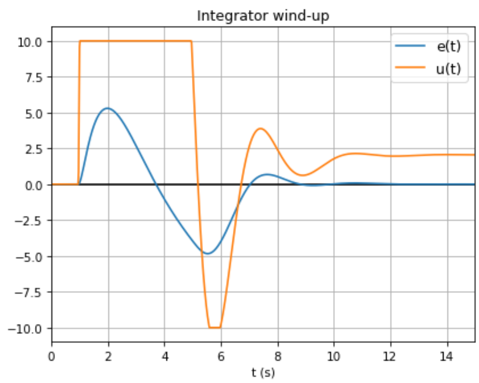


Figure 14: Integrator wind-up: large overshoot.

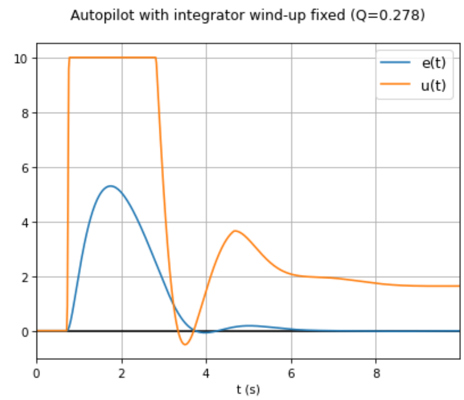


Figure 15: Wind-up fixed with $Q = 0.278$.